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Finite Deflection Structural Analysis Using Plate and Shell Discrete Elements

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An analysis capability for predicting the nonlinear behavior of elastic structural systems that can be adequately idealized with rectangular plate and cylindrical shell discrete elements is reported. The displacement patterns in each element are approximated in terms of products of one-dimensional, first-order Hermite interpolation polynomials and undetermined nodal displacement parameters. Geometric admissibility of the displacement state for an assembled set of these discrete elements can be readily assured. Incorporation of geometric nonlinearities in the strain-displacement relations permits the prediction of finite displacements. For example, elastic postbuckling behavior can be analyzed using this method. The option to impose a prescribed end-shortening and then determine the boundary-force distribution required to maintain the resulting displacement pattern is a valuable feature of this capability. Numerical solutions are obtained by direct minimization of the total potential energy. The elastic postbuckling behavior of a flat-plate panel and a cylindrical shell panel are presented as examples. The applicability of these discrete elements to predicting the elastic postbuckling behavior of integrally stiffened plate and cylindrical shell panels is provided by the assumed element displacement patterns.

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Introduction

THE quest for minimum-weight optimum structural designs in aerospace applications inherently tends to undercut the validity of the assumptions leading to linear formulations of the structural analysis problem. In order to predict structural behavior, it has frequently become necessary to base equilibrium considerations on deformed geometry, to employ more exact deformation-displacement relations and to consider the nonlinear behavior of materials.

The analysis method presented in this paper seeks to improve the prediction of structural behavior by taking into account geometric nonlinearities, while retaining the flexibility of application characteristic of discrete element structural analysis methods. The principal limitations on the method presented are that the rotations of the deformed structure relative to the undeformed position of the structure must be small (i.e., $\sin\theta \approx \theta$ and $\cos\theta \approx 1$) and the material behavior is assumed to remain linearly elastic. Extension of this approach to a material behavior law which can be represented by a strain energy density type of potential function is straightforward in principle; however, its implementation would require substantial effort.

The discrete element displacement method of structural analysis for linear structural systems is well-established. Previous extensions of the method to include geometric nonlinearity may be divided into three categories. The first category includes the approaches that take geometric nonlinearity into account by solving a sequence of linear problems. Procedures in this category are characterized by incremental application of the loading, the use of stiffness matrices exhibiting the influence of initial forces, and updating of the nodal positions (for example, see Refs. 1–3). The second category embraces those approaches that take geometric nonlinearity into account by formulating the set of nonlinear simultaneous algebraic equations governing the behavior of the structural systems and then seeking to solve them by successive approximation (for example, see Refs. 4 and 5). Finally, the third category consists of those approaches that take geometric nonlinearity into account by using nonlinear strain-displacement equations to construct the potential energy for each of the discrete elements, and then obtaining a numerical solution by seeking the minimum of the total potential energy for the assemblage of discrete elements representing the structure (for example, see Refs. 6–8).

Two of the previously cited references deal with geometric nonlinearity in displacement methods for plate-type elements wherein bending-membrane action is coupled. The approach used in Ref. 3 depended upon a sequence of linear solutions (load incrementations) and the displacement patterns assumed in each element for the displacement components u, v, and w were such that the interelement geometric admissibility conditions were not satisfied. In Ref. 6, the displacement patterns were such that the geometric admissibility conditions were satisfied everywhere. The assumed displacement pattern employed for the w displacement component was, however, incomplete.[9] In Ref. 10 the products of one-dimensional Hermite interpolation polynomials were used as assumed element displacement patterns to generate stiffness and mass matrices for rectangular plate bending elements. These elements readily lend themselves to the satisfaction of geometric admissibility conditions over an entire assemblage of finite elements representing a structure. The 16-degree-of-freedom rectangular plate bending element presented in the Addendum of Ref. 10 has been proven to satisfy the completeness condition in Ref. 9 and it has also been shown[11] that it is an adequate representation of the continuum with an error of the order of the mesh size squared.

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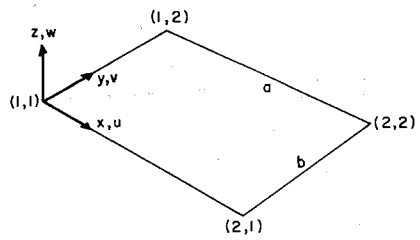


Fig. 1 Typical rectangular plate element.

The results reported here evolved from an investigation that began with the generation of a geometrically nonlinear bending-membrane rectangular plate element using bilinear interpolation of the inplane displacements u, v and bicubic interpolation of the transverse displacement w. This 24-degree-of-freedom ($4u$, $4v$, and $16w$) geometrically nonlinear element has been used to obtain solutions for various finite deflection plate problems with satisfactory results.

Rather than attempting to model cylindrical-shell structures with plate elements, effort was directed toward generating stiffness and mass matrices for a linear cylindrical shell element using the same level of interpolation employed in the nonlinear bending-membrane plate (i.e., bilinear interpolation for u and v and bicubic interpolation for w). The idealization of a curved structure using flat elements is not intuitively appealing. In Ref. 11 it is shown that flat elements do not adequately represent the bending behavior of a curved structure. The use of plate elements to represent shell structures appears to be particularly difficult to justify if buckling and/or postbuckling behavior is of interest.

An eigenvalue analysis of the 24 × 24 stiffness matrix obtained for the linear cylindrical shell element ($4u$, $4v$, $16w$ degrees of freedom) disclosed that the assumed displacement patterns admit to only four linearly independent displacement states that produce essentially zero strain energy compared with the other states.§ This deficiency led to the use of bicubic interpolation for all three displacement components. The stiffness and mass matrices for this 48-degree-of-freedom ($16u$, $16v$, and $16w$) cylindrical shell element based on linear strain-displacement equations are reported in Ref. 12. An eigenvalue analysis of this 48 × 48 stiffness matrix indicated that the bicubic assumed displacement patterns admit six linearly independent displacement states that produce "very little" strain energy compared with the other states. (See Appendix A for further remarks on the rigid-body mode problem.)

It should be noted that the linear, rectangular, flat-plate bending element with 16 degrees-of-freedom (Addendum, Ref. 10) and a linear, rectangular, membrane element with 32 degrees of freedom are special cases of the 48-degree-of-freedom cylindrical shell discrete element.¶

This paper reports the successful generation of a cylindrical shell element and a rectangular, flat-plate element based on finite-deflection, strain-displacement relations. The displacement patterns for u, v, and w in both of these discrete elements are approximated in terms of products of one-dimensional, "cubic Hermite," interpolation polynomials producing 48-degree-of-freedom elements. It should be noted that although the use of these interpolation formulas for the membrane displacement pattern increases the number of degrees of freedom, it 1) leads to adequate representation of the six rigid-body modes, 2) makes it possible to join these plate and cylindrical shell elements together at any angle, and 3) provides for a more accurate description of the membrane stress state which can be particularly valuable in predicting buckling and postbuckling behavior.

Formulation

In this section, mathematical models which facilitate the prediction of the nonlinear behavior of elastic structural systems are formulated for a class of structures composed of rectangular plate and cylindrical shell substructures. These models are given in the form of total, discretized potential energy functions obtained by summing the contributions of the discrete elements which comprise the idealized system. Expressions for the strain energy in a convenient matrix form are developed for typical rectangular plate and cylindrical shell discrete elements incorporating geometric nonlinearities through the strain-displacement relations. Geometric admissibility of the displacement pattern over an assemblage of discrete elements is facilitated by expressing the assumed displacement shape for a typical element in terms of the interpolation formulas.

Displacement Modes

The middle surface displacements u, v, and w of a typical discrete element are approximated by a sum of products of one-dimensional, cubic interpolation polynomials and undetermined parameters. For example, for a rectangular plate element,

$$w(x,y) = \sum_{i=1}^{2}\sum_{j=1}^{2} [H_{0i}^{(1)}(x)H_{0j}^{(1)}(y)\alpha_{ij} + H_{1i}^{(1)}(x)H_{0j}^{(1)}(y) \times \beta_{ij} + H_{0i}^{(1)}(x)H_{1j}^{(1)}(y)\gamma_{ij} + H_{1i}^{(1)}(x)H_{1j}^{(1)}(y)\eta_{ij}] \quad (1)$$

where particular values of the indices i and j refer to a node of the element (Fig. 1); α_{ij}, β_{ij}, γ_{ij}, and η_{ij} are undetermined parameters; and $H_{ki}^{(1)}$ are the Hermite interpolation polynomials and are given by**

$$H_{01}^{(1)}(x) = (2x^3 - 3a\,x^2 + a^3)/a^3$$
$$H_{02}^{(1)}(x) = -(2x^3 - 3a\,x^2)/a^3$$
$$H_{11}^{(1)}(x) = (x^3 - 2a\,x^2 + a^2x)/a^2$$
$$H_{12}^{(1)}(x) = (x^3 - ax^2)/a^2$$

Similar expressions for the y-direction are obtained by replacing x by y and a by b. The aforementioned polynomials have the following properties:

$$H_{0i}^{(1)}(x_k) = \delta_{ik} \qquad H_{1i}^{(1)}(x_k) = 0$$
$$dH_{0i}^{(1)}(x_k)/dx = 0 \qquad dH_{1i}^{(1)}(x_k)/dx = \delta_{ik} \quad (2)$$

When $w(x,y)$, as given by Eq. (1), and its derivatives are evaluated at the four nodes of the element and the aforesaid properties are invoked, it is observed that the 16 undetermined parameters α_{ij}, β_{ij}, γ_{ij}, and η_{ij} are exactly the 16 nodal quantities w_{ij}, w_{xij}, w_{yij}, and w_{xyij}, respectively, where the subscripts x and y denote partial differentiation. The plate element assumed displacement state then takes the form

$$w(x,y) = \sum_{i=1}^{2}\sum_{j=1}^{2} [H_{0i}^{(1)}(x)H_{0j}^{(1)}(y)w_{ij} + H_{1i}^{(1)}(x)H_{0j}^{(1)}(y) \times w_{xij} + H_{0i}^{(1)}(x)H_{1j}^{(1)}(y)w_{yij} + H_{1i}^{(1)}(x)H_{1j}^{(1)}(y)w_{xyij}] \quad (3)$$

§ The zero strain energy displacement states are often called the rigid-body modes. However, it should be understood that these rigid-body motions are approximations to the general rigid-body displacements of kinematics which take the form of a general orthogonal transformation and a translation (see Appendix A).

¶ Explicit formulas for the elements of these stiffness and mass matrices can be obtained from the expressions given in Ref. 12 by replacing $r\Delta\theta$ with b, l with a, and $1/r$ with zero.

** The parenthetical superscript (1) in $H_{ki}^{(1)}$ is part of a general notation system adopted by the authors. The work reported here is limited to two point interpolation formulas and the superscript refers to the highest derivative interpolated at the end points. In this case the (1) indicates that the function and its first derivative (slope) are matched.

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with similar expressions for the u and v displacements. In Ref. 10, this form for the assumed w displacement is used to generate linear stiffness and mass matrices for a rectangular plate bending element having 16 degrees of freedom.

The displacements for a cylindrical-shell discrete element (Fig. 2) are given by approximations similar to Eq. (3) by replacing x by s, a by l, y by $r\theta$, and b by $r\Delta\theta$. Linear stiffness and mass matrices are given in Ref. 12 using these assumed displacement expressions.

It is noted at this juncture that it is the form of the assumed displacement modes of a typical discrete element [Eq. (3)] and the properties of the interpolation polynomials [Eq. (2)] which make it possible to readily satisfy geometric admissibility of the displacement pattern for an assembly of elements. A more complete discussion of the satisfaction of the admissibility requirements is given in the section entitled "Implementation."

Potential Energy

The strain energy of an isotropic, linearly elastic, thin rectangular plate element of thickness h is given by

$$U_p = \int_0^a \int_0^b \int_{-h/2}^{h/2} \frac{E}{2(1-\nu^2)} \times [\epsilon_x^2 + \epsilon_y^2 + 2\nu\epsilon_x\epsilon_y + (1-\nu)\epsilon_{xy}^2/2]\, dzdydx \quad (4)$$

and for a cylindrical shell element by

$$U_c = \int_0^l \int_0^{\Delta\theta} \int_{-h/2}^{h/2} \frac{E}{2(1-\nu^2)} \times [\epsilon_s^2 + \epsilon_\theta^2 + 2\nu\epsilon_s\epsilon_\theta + (1-\nu)\epsilon_{s\theta}^2/2]\, rdzd\theta ds \quad (5)$$

The finite-displacement, strain-displacement relations are, for the plate,[13]

$$\begin{aligned} \epsilon_x &= (\partial u/\partial x) + \tfrac{1}{2}(\partial w/\partial x)^2 - z(\partial^2 w/\partial x^2) \\ \epsilon_y &= (\partial v/\partial y) + \tfrac{1}{2}(\partial w/\partial y)^2 - z(\partial^2 w/\partial y^2) \\ \gamma_{xy} &= \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial w}{\partial x}\frac{\partial w}{\partial y} - 2z\frac{\partial^2 w}{\partial x \partial y} \end{aligned} \quad (6)$$

and for the cylinder,[14]

$$\begin{aligned} \epsilon_s &= (\partial u/\partial s) + \tfrac{1}{2}(\partial w/\partial s)^2 - z(\partial^2 w/\partial s^2) \\ \epsilon_\theta &= \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{w}{r} + \frac{1}{2r^2}\left(\frac{\partial w}{\partial \theta}\right)^2 - \frac{z}{r^2}\left(\frac{\partial^2 w}{\partial \theta^2} - \frac{\partial v}{\partial \theta}\right) \\ \gamma_{s\theta} &= \frac{\partial v}{\partial s} + \frac{1}{r}\frac{\partial u}{\partial \theta} + \frac{1}{r}\frac{\partial w}{\partial s}\frac{\partial w}{\partial \theta} - \frac{2z}{r}\left(\frac{\partial^2 w}{\partial s \partial \theta} - \frac{\partial v}{\partial s}\right) \end{aligned} \quad (7)$$

Upon substitution of the strain-displacement relations [Eqs. (6) and (7)] and integration through the thickness, the strain energy expressions [Eqs. (4) and (5)] become, for the plate element,

$$U_p = U_{2p} + U_{3p} + U_{4p} \quad (8)$$

where (the subscripts 2, 3, and 4 refer to quadratic, cubic and fourth-order terms, respectively),

$$U_{2p} = \frac{D}{2}\int_0^a\int_0^b \left\{\frac{12}{h^2}\left[\left(\frac{\partial u}{\partial x}\right)^2 + \frac{1-\nu}{2}\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \frac{1-\nu}{2}\left(\frac{\partial v}{\partial x}\right)^2\right] + \left(\frac{\partial^2 w}{\partial x^2}\right)^2 + \left(\frac{\partial^2 w}{\partial y^2}\right)^2 + 2\nu\frac{\partial^2 w}{\partial x^2}\frac{\partial^2 w}{\partial y^2} + 2(1-\nu)\left(\frac{\partial^2 w}{\partial x \partial y}\right)^2 + \frac{12}{h^2}\left[2\nu\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} + (1-\nu)\frac{\partial u}{\partial y}\frac{\partial v}{\partial x}\right]\right\} dydx \quad (8a)$$

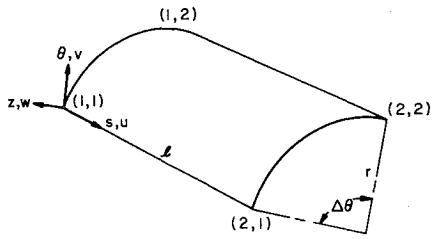


Fig. 2 Typical cylindrical element.

$$U_{3p} = \frac{D}{2}\int_0^a\int_0^b \frac{12}{h^2}\left\{\frac{\partial u}{\partial x}\left[\left(\frac{\partial w}{\partial x}\right)^2 + \nu\left(\frac{\partial w}{\partial y}\right)^2\right] + \frac{\partial v}{\partial y}\left[\left(\frac{\partial w}{\partial y}\right)^2 + \nu\left(\frac{\partial w}{\partial x}\right)^2\right] + (1-\nu)\times\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\frac{\partial w}{\partial x}\frac{\partial w}{\partial y}\right\} dydx \quad (8b)$$

$$U_{4p} = \frac{D}{2}\int_0^a\int_0^b \frac{12}{h^2}\left[\frac{1}{4}\left(\frac{\partial w}{\partial x}\right)^4 + \frac{1}{4}\left(\frac{\partial w}{\partial y}\right)^4 + \frac{1}{2}\left(\frac{\partial w}{\partial x}\right)^2\left(\frac{\partial w}{\partial y}\right)^2\right] dydx \quad (8c)$$

and for the cylinder element,

$$U_c = U_{2c} + U_{3c} + U_{4c} \quad (9)$$

where,

$$U_{2c} = \frac{D}{2}\int_0^l\int_0^{\Delta\theta} \left\{\frac{12}{h^2}\left[\left(\frac{\partial u}{\partial s}\right)^2 + \frac{(1-\nu)}{2r^2}\left(\frac{\partial u}{\partial \theta}\right)^2\right] + \frac{1}{r^2}\left(\frac{12}{h^2} + \frac{1}{r^2}\right)\left(\frac{\partial v}{\partial \theta}\right)^2 + 2(1-\nu)\left(\frac{3}{h^2} + \frac{1}{r^2}\right)\left(\frac{\partial v}{\partial s}\right)^2 + \left(\frac{\partial^2 w}{\partial s^2}\right)^2 + \frac{1}{r^4}\left(\frac{\partial^2 w}{\partial \theta^2}\right)^2 + \frac{2\nu}{r^2}\frac{\partial^2 w}{\partial s^2}\frac{\partial^2 w}{\partial \theta^2} + \frac{2(1-\nu)}{r^2}\left(\frac{\partial^2 w}{\partial s \partial \theta}\right)^2 + \frac{12}{h^2 r^2}w^2 + \frac{12}{h^2}\left[\frac{2\nu}{r}\frac{\partial u}{\partial s}\frac{\partial v}{\partial \theta} + \frac{1-\nu}{r}\frac{\partial u}{\partial \theta}\frac{\partial v}{\partial s}\right] + \frac{24}{h^2 r^2}\left[\nu\frac{\partial u}{\partial s} + \frac{\partial v}{\partial \theta}\right]w - \frac{2}{r^2}\frac{\partial v}{\partial \theta}\times\left[\frac{1}{r^2}\frac{\partial^2 w}{\partial \theta^2} + \nu\frac{\partial^2 w}{\partial s^2}\right] - \frac{4(1-\nu)}{r^2}\frac{\partial v}{\partial s}\frac{\partial^2 w}{\partial s \partial \theta}\right\} rd\theta ds \quad (9a)$$

$$U_{3c} = \frac{D}{2}\int_0^l\int_0^{\Delta\theta} \frac{12}{h^2}\left\{\frac{\partial u}{\partial s}\left[\left(\frac{\partial w}{\partial s}\right)^2 + \frac{\nu}{r^2}\left(\frac{\partial w}{\partial \theta}\right)^2\right] + \frac{1}{r}\frac{\partial v}{\partial \theta}\left[\frac{1}{r^2}\left(\frac{\partial w}{\partial \theta}\right)^2 + \nu\left(\frac{\partial w}{\partial s}\right)^2\right] + \frac{(1-\nu)}{r}\times\left(\frac{1}{r}\frac{\partial u}{\partial \theta} + \frac{\partial v}{\partial s}\right)\frac{\partial w}{\partial s}\frac{\partial w}{\partial \theta} + \frac{1}{r^3}w\times\left(\frac{\partial w}{\partial \theta}\right)^2 + \frac{\nu}{r}w\left(\frac{\partial w}{\partial s}\right)^2\right\} rd\theta ds \quad (9b)$$

$$U_{4c} = \frac{D}{2}\int_0^l\int_0^{\Delta\theta} \frac{12}{h^2}\left[\frac{1}{4}\left(\frac{\partial w}{\partial s}\right)^4 + \frac{1}{4r^4}\left(\frac{\partial w}{\partial \theta}\right)^4 + \frac{1}{2r^2}\left(\frac{\partial w}{\partial s}\right)^2\left(\frac{\partial w}{\partial \theta}\right)^2\right] rd\theta ds \quad (9c)$$

where D is the flexural rigidity $Eh^3/12(1-\nu^2)$. The external work done by the applied loads p_u, p_v, and p_w acting on a typical plate or cylinder element in the u, v, and w directions, respectively, is given by

$$W = \int_A [p_u u + p_v v + p_w w]dA \quad (10)$$

Upon substitution of the assumed displacement patterns [Eq. (3)] into Eqs. (8) and (9) and performing the indicated

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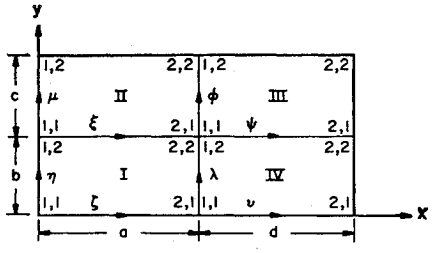


Fig. 3 Four adjacent elements.

integrations over the middle surface the discretized strain energy for a plate element can be expressed compactly by

$$U_{2p} = \underset{1\times48}{\tfrac{1}{2}X_p^T}\ \underset{48\times48}{K_{2p}}\ \underset{48\times1}{X_p} \tag{11a}$$

$$U_{3p} = \underset{1\times48}{\tfrac{1}{2}X_p^T}\ \underset{48\times136}{K_{3p}}\ \underset{136\times1}{Y_p} \tag{11b}$$

$$U_{4p} = \underset{1\times136}{\tfrac{1}{8}Y_p^T}\ \underset{136\times136}{K_{4p}}\ \underset{136\times1}{Y_p} \tag{11c}$$

and analogously for a cylinder by replacing p with c,

$$U_{2c} = \underset{1\times48}{\tfrac{1}{2}X_c^T}\ \underset{48\times48}{K_{2c}}\ \underset{48\times1}{X_c} \tag{11d}$$

$$U_{3c} = \underset{1\times48}{\tfrac{1}{2}X_c^T}\ \underset{48\times136}{K_{3c}}\ \underset{136\times1}{Y_c} \tag{11e}$$

$$U_{4c} = \underset{1\times136}{\tfrac{1}{8}Y_c^T}\ \underset{136\times136}{K_{4c}}\ \underset{136\times1}{Y_c} \tag{11f}$$

The vector X_p (or X_c) contains the 48 nodal degrees of freedom of a plate (or cylinder) element; the vector Y_p (or Y_c) contains the 136 possible combinations of the $16w$ degrees of freedom; K_{2p} (or K_{2c}) is the ordinary stiffness matrix associated with the quadratic portion of the strain energy; K_{3p} (or K_{3c}) and K_{4p} (or K_{4c}) are matrices resulting from the cubic and fourth-order energy terms, respectively. (Refer to Appendix B for a more complete development of the discretized strain energy.)

The discretized work term [Eq. (10)] is written as

$$W_p = \underset{1\times48}{X_p^T}\ \underset{48\times1}{P_p} \tag{12a}$$

or

$$W_c = \underset{1\times48}{X_c^T}\ \underset{48\times1}{P_c} \tag{12b}$$

where P_p (or P_c) is a vector containing the 48 work equivalent loads associated with each of the 48 nodal degrees of freedom of a plate (cylinder) element. The potential energy of a rectangular plate or cylindrical shell discrete element is given by the well-known expression

$$\pi_{p_p} = U_p - W_p \tag{13a}$$

or

$$\pi_{p_c} = U_c - W_c \tag{13b}$$

Finally, the total potential energy of an assemblage of N_p plate elements and N_c cylinder elements is written as

$$\Pi_p = \sum_{k=1}^{N_p} \pi_{p_p}{}^{(k)} + \sum_{k=1}^{N_c} \pi_{p_c} \tag{14}$$

Implementation

The total potential energy for an assemblage of elements representing a structure [Eq. (14)] must be expressed in terms of independent degrees of freedom by satisfying the conditions of geometric admissibility prior to seeking an approximate solution by finding the minimum of the total potential energy.

Interelement Geometric Admissibility

To illustrate how the geometric admissibility conditions between interior finite elements are satisfied, consider the group of four rectangular elements shown in Fig. 3. For example, the geometric admissibility conditions on the interface between elements I and II are

$$u^{\mathrm{I}}(\zeta,b) = u^{\mathrm{II}}(\xi,0) \tag{15a}$$

$$v^{\mathrm{I}}(\zeta,b) = v^{\mathrm{II}}(\xi,0) \tag{15b}$$

$$w^{\mathrm{I}}(\zeta,b) = w^{\mathrm{II}}(\xi,0) \tag{15c}$$

$$\partial w^{\mathrm{I}}(\zeta,b)/\partial\eta = \partial w^{\mathrm{II}}(\xi,0)/\partial\mu \tag{15d}$$

where 1) $w^{\mathrm{I}}(\zeta,b)$ = the w displacement of element I along the edge $\eta = b$ where ζ and η are local coordinates of element I with origin at the corner denoted 1,1 in element I; 2) $w^{\mathrm{II}}(\xi,0)$ = the w displacement of element II along the edge $\mu = 0$ where ξ and μ are local coordinates of element II with origin at the corner denoted 1,1 in element II, with corresponding definitions for the u, v, $\partial w/\partial\eta$ and $\partial w/\partial\mu$ terms. It is evident that the distinction between ξ and ζ along the interface can be dropped and in what follows the identity and definition $\xi = \zeta \equiv x$ will be employed. From Eq. (3) and the properties of the first-order Hermite interpolation polynomials [Eq. (2)] it follows that the displacements and their first derivatives on an edge will depend only upon nodal quantities on that same edge. For example,

$$w^{\mathrm{I}}(x,b) = \sum_{i=1}^{2} [H_{0i}{}^{(1)}(x)\, w_{i2}{}^{\mathrm{I}} + H_{1i}{}^{(1)}(x)\, w_{xi2}{}^{\mathrm{I}}] \tag{16}$$

and

$$w^{\mathrm{II}}(x,0) = \sum_{i=1}^{2} [H_{0i}{}^{(1)}(x) w_{i1}{}^{\mathrm{II}} + H_{1i}{}^{(1)}(x) w_{xi1}{}^{\mathrm{II}}] \tag{17}$$

where the Roman superscripts on the undetermined nodal quantities w_{ij}, w_{xij}, etc., identify the discrete element with which they are associated. Substituting Eqs. (16) and (17) into Eq. (15c) and noting that $H_{0i}{}^{(1)}(x)$ and $H_{1i}{}^{(1)}(x)$ are linearly independent, it follows that the geometric admissibility condition represented by Eq. (15c) is satisfied along the entire edge, provided

$$w_{i2}{}^{\mathrm{I}} = w_{i1}{}^{\mathrm{II}} \quad \text{and} \quad w_{xi2}{}^{\mathrm{I}} = w_{xi1}{}^{\mathrm{II}} \quad i = 1,2 \tag{18}$$

From Eqs. (2) and (3), it also follows that

$$\frac{\partial w^{\mathrm{I}}(x,b)}{\partial y} = \sum_{i=1}^{2} [H_{0i}{}^{(1)}(x) w_{yi2}{}^{\mathrm{I}} + H_{1i}{}^{(1)}(x) w_{xyi2}{}^{\mathrm{I}}] \tag{19}$$

and

$$\frac{\partial w^{\mathrm{II}}(x,0)}{\partial y} = \sum_{i=1}^{2} [H_{0i}{}^{(1)}(x) w_{yi1}{}^{\mathrm{II}} + H_{1i}{}^{(1)}(x) w_{xyi1}{}^{\mathrm{II}}] \tag{20}$$

Substituting Eqs. (19) and (20) into Eq. (15d), it follows that this geometric admissibility condition is satisfied along the common edge provided

$$w_{yi2}{}^{\mathrm{I}} = w_{yi1}{}^{\mathrm{II}} \quad \text{and} \quad w_{xyi2}{}^{\mathrm{I}} = w_{xyi1}{}^{\mathrm{II}} \quad i = 1,2 \tag{21}$$

In a similar manner, the geometric admissibility conditions represented by Eqs. (15a) and (15b), respectively, are satisfied along the common edge provided

$$u_{i2}{}^{\mathrm{I}} = u_{i1}{}^{\mathrm{II}} \qquad u_{xi2}{}^{\mathrm{I}} = u_{xi1}{}^{\mathrm{II}} \qquad i = 1,2 \tag{22}$$

and

$$v_{i2}{}^{\mathrm{I}} = v_{i1}{}^{\mathrm{II}} \qquad v_{xi2}{}^{\mathrm{I}} = v_{xi1}{}^{\mathrm{II}} \qquad i = 1,2 \tag{23}$$

The same procedure can be followed to determine the linking conditions required to satisfy the geometric admissibility

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conditions at the interfaces between elements II-III, III-IV, and IV-I. Prior to imposing the geometric admissibility conditions, the number of independent degrees of freedom at the interior node point common to discrete elements I, II, III, and IV is 48, namely 12 for each element (u^{I}, $u_x{}^{\mathrm{I}}$, $u_y{}^{\mathrm{I}}$, $u_{xy}{}^{\mathrm{I}}$, v^{I}, $v_x{}^{\mathrm{I}}$, $v_y{}^{\mathrm{I}}$, $v_{xy}{}^{\mathrm{I}}$, w^{I}, $w_x{}^{\mathrm{I}}$, $w_y{}^{\mathrm{I}}$, $w_{xy}{}^{\mathrm{I}}$; u^{II}, . . . , $w_{xy}{}^{\mathrm{II}}$; u^{III}, . . . , $w_{xy}{}^{\mathrm{III}}$; and u^{IV}, . . . , $w_{xy}{}^{\mathrm{IV}}$). The requirements at the interior node point common to elements I, II, III, and IV that result from satisfying the geometric admissibility conditions along the interfaces between elements I, II, III, and IV can be summarized as follows:

$$w_{22}{}^{\mathrm{I}} = w_{21}{}^{\mathrm{II}} = w_{11}{}^{\mathrm{III}} = w_{12}{}^{\mathrm{IV}} \quad (24a)$$

$$w_{x22}{}^{\mathrm{I}} = w_{x21}{}^{\mathrm{II}} = w_{x11}{}^{\mathrm{III}} = w_{x12}{}^{\mathrm{IV}} \quad (24b)$$

$$w_{y22}{}^{\mathrm{I}} = w_{y21}{}^{\mathrm{II}} = w_{y11}{}^{\mathrm{III}} = w_{y12}{}^{\mathrm{IV}} \quad (24c)$$

$$w_{xy22}{}^{\mathrm{I}} = w_{xy21}{}^{\mathrm{II}} = w_{xy11}{}^{\mathrm{III}} = w_{xy12}{}^{\mathrm{IV}} \quad (24d)$$

$$u_{22}{}^{\mathrm{I}} = u_{21}{}^{\mathrm{II}} = u_{11}{}^{\mathrm{III}} = u_{12}{}^{\mathrm{IV}} \quad (24e)$$

$$u_{x22}{}^{\mathrm{I}} = u_{x21}{}^{\mathrm{II}}; \quad u_{x11}{}^{\mathrm{III}} = u_{x12}{}^{\mathrm{IV}} \quad (24f)$$

$$u_{y21}{}^{\mathrm{II}} = u_{y11}{}^{\mathrm{III}}; \quad u_{y12}{}^{\mathrm{IV}} = u_{y22}{}^{\mathrm{I}} \quad (24g)$$

$$v_{22}{}^{\mathrm{I}} = v_{21}{}^{\mathrm{II}} = v_{11}{}^{\mathrm{III}} = v_{12}{}^{\mathrm{IV}} \quad (24h)$$

$$v_{x22}{}^{\mathrm{I}} = v_{x21}{}^{\mathrm{II}}; \quad v_{x11}{}^{\mathrm{III}} = v_{x12}{}^{\mathrm{IV}} \quad (24i)$$

$$v_{y21}{}^{\mathrm{II}} = v_{y11}{}^{\mathrm{III}}; \quad v_{y12}{}^{\mathrm{IV}} = v_{y22}{}^{\mathrm{I}} \quad (24j)$$

Thus, imposing the geometric admissibility conditions along the common edges of the four elements reduces the number of independent degrees of freedom at a typical interior node point to 22.

Additional Linking Conditions

The foregoing discussion explains how the imposing of geometric admissibility conditions reduces the number of degrees of freedom. It should, however, be noted that for problems where the engineer has some further insight into the expected behavior of the structure, further reduction in the number of independent degrees of freedom can frequently be achieved. For example, let us suppose that it is anticipated that the normal and shear strains (ϵ_x, ϵ_y, γ_{xy}) as well as the rotation $\{\theta_z = \frac{1}{2}[(\partial v/\partial x) - (\partial u/\partial y)]\}$ will be continuous at the interfaces between the elements I, II, III, and IV depicted in Fig. 3. On the interface between elements I and II, this requirement is equivalent to (since u, v, w, $\partial w/\partial y$ have previously been made continuous)

$$\partial u^{\mathrm{I}}(\zeta,b)/\partial\eta = \partial u^{\mathrm{II}}(\xi,0)/\partial\mu \quad (25a)$$

$$\partial v^{\mathrm{I}}(\zeta,b)/\partial\eta = \partial v^{\mathrm{II}}(\xi,0)/\partial\mu \quad (25b)$$

and the new linking conditions needed to satisfy Eqs. (25a) and (25b) can be summarized by simply replacing the semicolon in Eqs. (24f, g, i, and j) and adding the conditions

$$u_{xy}{}^{\mathrm{I}} = u_{xy}{}^{\mathrm{II}} = u_{xy}{}^{\mathrm{III}} = u_{xy}{}^{\mathrm{IV}} \quad (26a)$$

$$v_{xy}{}^{\mathrm{I}} = v_{xy}{}^{\mathrm{II}} = v_{xy}{}^{\mathrm{III}} = v_{xy}{}^{\mathrm{IV}} \quad (26b)$$

Imposing the additional requirements that the strains ϵ_x, ϵ_y, γ_{xy} and the rotation θ_z be continuous along the interfaces between elements I, II, III, and IV reduces the number of independent degrees of freedom at the interior node point to 12 (namely, u, u_x, u_y, u_{xy}, v, v_x, v_y, v_{xy}, w, w_x, w_y, and w_{xy}). It should be pointed out that although additional linking conditions are optional, employing them when appropriate reduces the number of degrees of freedom without a substantial change in the results. All of the numerical results reported in the illustrative examples section of this paper were obtained by employing these additional linking conditions.

Boundary Conditions

Imposing the prescribed displacement boundary conditions will lead to a further reduction in the number of degrees of freedom. For example, suppose the edge $\eta = 0$ of element I in Fig. 3 is clamped. Then the displacement boundary conditions to be imposed are

$$w^{\mathrm{I}}(x,0) = \sum_{i=1}^{2} [H_{0i}{}^{(1)}(x)w_{i1}{}^{\mathrm{I}} + H_{1i}{}^{(1)}(x)w_{xi1}{}^{\mathrm{I}}] = 0 \quad (27)$$

and

$$\frac{\partial w^{\mathrm{I}}(x,0)}{\partial y} = \sum_{i=1}^{2} [H_{0i}{}^{(1)}(x)w_{yi1}{}^{\mathrm{I}} + H_{1i}{}^{(1)}(x)w_{xyi1}{}^{\mathrm{I}}] = 0 \quad (28)$$

which imply the requirements that

$$w_{i1}{}^{\mathrm{I}} = w_{xi1}{}^{\mathrm{I}} = w_{yi1}{}^{\mathrm{I}} = w_{xyi1}{}^{\mathrm{I}} = 0 \qquad i = 1,2 \quad (29)$$

Force boundary conditions are treated as applied loads, and they are dealt with using the work equivalence approach. For example, suppose a given in-plane load distribution $\bar{N}_y(x)$ is applied along the edge $\mu = b$ of element II (see Fig. 3) then the contribution to the work expression $W_p = \mathbf{X}_p{}^T P_p$ [Eq. (12a)] for plate element II is given by

$$\int_0^a \bar{N}_y(x)v(x,b)dx = \sum_{i=1}^{2} \left[v_{i2}{}^{\mathrm{I}} \int_0^a \bar{N}_y(x)H_{0i}{}^{(1)}(x)dx + v_{xi2}{}^{\mathrm{I}} \int_0^a \bar{N}_y(x)H_{1i}{}^{(1)}(x)dx \right] \quad (30)$$

It should be noted that since the principle of minimum potential energy is taken as the basis for obtaining approximate solutions, force boundary conditions are natural, whereas displacement boundary conditions are imposed in the variational sense. Therefore, prior to seeking an approximate solution by minimizing the total potential energy, only the displacement boundary conditions must be satisfied by the assumed displacement state as a part of the geometric admissibility requirement.

Variable Correlation Table

The preceding discussion of geometric admissibility, additional conditions, and boundary conditions provides a means for deciding what the independent degrees of freedom are for a particular structure idealized as an assemblage of discrete elements. Each of these degrees of freedom is assigned a number from 1 to n (the total number of independent degrees of freedom) and a corresponding location in an n-component vector q. The elements of this vector are viewed as the coordinates of an n-dimensional space in which the total potential energy is to be minimized. The potential energy is, however, calculated according to Eq. (14) where each of the element potential energies is a function of its 48 nodal displacement parameters which are, in general, not independent. Therefore, some procedure for correlating the nodal parameters of the elements with the components of the vector q must be provided so that the potential energy can be minimized with respect to the independent degrees of freedom but calculated as a function of the element nodal parameters.

A convenient means by which the independent degrees of freedom may be correlated with the element nodal parameters is through a "variable correlation table." The variable correlation table is an array each row of which corresponds to a particular element in an assemblage, and each column of which corresponds to one of the 48 nodal parameters of an element [the nodal parameters are ordered according to Eqs. (B1) and (B2)]. The elements of the variable correlation table, in effect, describe the imposed displacement boundary conditions and the previously discussed linking conditions between adjacent discrete elements as follows.

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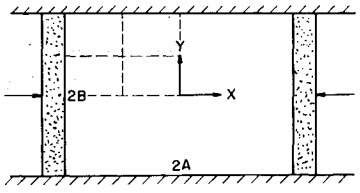


Fig. 4 Illustrative example.

Let VC_{ij} denote a particular element of the variable correlation table, where j represents one of the 48 nodal parameters of element number i. If the nodal parameter in question is restricted to be zero by an imposed displacement boundary condition, then VC_{ij} is assigned the number 0; if this nodal parameter has been linked to an independent degree of freedom, then VC_{ij} is assigned the number (from 1 to n) that corresponds to the location of this variable in the vector q; if this nodal parameter is opposite in sign to one of the independent degrees of freedom (this can happen if the local coordinates of adjacent elements are not in the same directions), then VC_{ij} is assigned the negative of the number which corresponds to the location of this variable in the vector q.

The correlation between the components of the vector of independent degrees of freedom q and the vector of nodal parameters for the ith discrete element $X^{(i)}$ is represented in equation form by

$$X_j^{(i)} = \begin{cases} \operatorname{sgn}(VC_{ij}) q_{|VC_{ij}|} & \text{if } VC_{ij} \neq 0 \\ 0 & \text{if } VC_{ij} = 0 \end{cases} \qquad j = 1, \ldots, 48 \tag{31}$$

The calculation of the gradient of the potential energy, $\nabla\pi_p^{(i)}$, with respect to the nodal parameters of the ith element in an assemblage is outlined in Appendix B [Eq. (B23)]. The gradient g of the total potential energy of the assemblage can be computed from the element gradients by use of the variable correlation table as follows: compute

$$g_{|VC_{ij}|}{}^{(i)} = \operatorname{sgn}(VC_{ij}) \nabla\pi_{pj}{}^{(i)} \qquad j = 1, \ldots, 48$$

for $VC_{ij} \neq 0$ and $i = 1, \ldots, N$ where N is the number of elements in the assemblage. Then,

$$g = \sum_{i=1}^{N} g^{(i)} \tag{32}$$

Function Minimization

The formulation of the discrete elemental structural analysis problem has been cast into a function of the independent degrees of freedom, $F(q) = \Pi_p$. One of the most powerful of the many techniques for locating the unconstrained minimum of a function of many variables is that presented in Ref. 15. This iterative procedure locates the minimum of a function as the limit of a sequence $q^0, q^1, q^2, \ldots$ which is proved to converge quadratically. The vector q^0 is an initial approximation to the minimum.

The ith iteration begins with the determination of a direction of travel s^i according to the equation $s^i = -H^i g^i$, where g^i is the gradient of the function at the currently occupied point q^i and H^i is a matrix which initially may be chosen to be any positive definite symmetric matrix. For convenience H^0 is often selected as the identity matrix, thus beginning the search process in the direction of steepest descent. The function is then minimized along the generated direction by choosing $\alpha^i > 0$ such that $F(q^i + \alpha^i s^i)$ is a minimum. The next element in the sequence is then $q^{i+1} = q^i + \alpha^i s^i$. Prior to beginning another iteration, the matrix H is modified to take into account local characteristics of the function in order to avoid the zigzag behavior common to many other minimization techniques, $H^{i+1} = H^i + \Delta H^i$ where,

$$\Delta H^i = \frac{\alpha^i s^i s^{i\,T}}{s^{i\,T}(g^{i+1} - g^i)} - \frac{H^i(g^{i+1} - g^i)(g^{i+1} - g^i)^T H^i}{(g^{i+1} - g^i)^T H^i (g^{i+1} - g^i)}$$

The modification of H is performed so that symmetry and positive definiteness are preserved, thus insuring stability of the procedure.

Illustrative Examples

The applicability of the proposed geometrically nonlinear plate and cylindrical shell discrete elements to the prediction of finite displacements and the postbuckling behavior of elastic structural systems is demonstrated through three numerical examples.

Flat Rectangular Plate

A simply supported rectangular plate subjected to in-plane compression is considered (Fig. 4). The compression is induced by imposing, through "rigid blocks," a uniform displacement of the edges $x = \pm A$. Lateral displacement of the edges $y = \pm B$, is prevented by constraining the plate between "walls." The resulting boundary conditions are

$$\begin{aligned} u(-A,y) &= \delta \\ u(A,y) &= -\delta \\ w(\pm A,y) &= w(x, \pm B) = 0 \\ v(\pm A,y) &= v(x, \pm B) = 0 \end{aligned} \tag{33}$$

The first two conditions imply that the edges $x = \pm A$ remain straight [i.e., $\partial u/\partial y\ (\pm A, y) = 0$].

Taking advantage of the double symmetry of the problem, one quadrant of the plate was modeled with four nonlinear, 48-degree-of-freedom plate discrete elements as shown in Fig. 4. Connecting appropriate degrees of freedom to insure a geometrically admissible assembled displacement pattern, requiring continuity of strains and taking the boundary conditions into account, the problem is reduced to the minimization of a function of 52 variables.

The load vs end-shortening plot is shown as the solid line in Fig. 5 for the particular case when $A = B = 12$ in., $h = 0.1$ in., $\nu = 0.3$, and $E = 30 \times 10^6$ psi. The ordinate N is the equivalent uniform edge load required to maintain the displacement pattern induced by the imposed end-shortening and is calculated by

$$N = \frac{Eh}{B(1 - \nu^2)} \int_0^B (\epsilon_x + \nu\epsilon_y)|_{x=-A}\, dy \tag{34}$$

The critical load predicted by minimization of the total discretized potential energy is

$$N_{cr} = 142.5 \text{ lb/in.}$$

as compared to

$$\bar{N}_{cr} = 0.632Eh^3/(1 - \nu^2)a^2 = 144.5 \text{ lb/in.}$$

given in Ref. 16.

The solid line in Fig. 6 represents the response of the transverse center deflection, w_0. The plot indicates that the plate remains flat until the critical load is attained. With increased loading into the postbuckled region, the center deflection

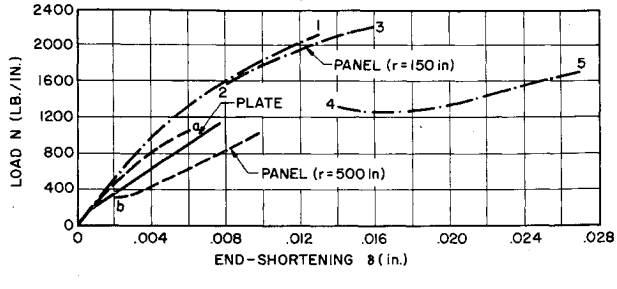


Fig. 5 Load vs end-shortening.

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increases uniformly. It is noted that the mirror image of this curve about the load axis is equally realizable.

Slightly Curved Panel

A slightly curved cylindrical shell panel subjected to in-plane compression is considered. As in the previous example, the compression is induced through rigid blocks so that a uniform end-shortening can be imposed (refer to Fig. 4 replacing x with s and y with $r\theta$). The panel is simply supported and circumferential displacement of the edges $r\theta = \pm B$ is prevented. The boundary conditions are expressed by Eq. (33) if x is replaced with s and y with $r\theta$.

Requiring symmetry about the two axes, one quadrant of the panel was modeled with four nonlinear, 48-degree-of-freedom cylindrical shell discrete elements as shown in Fig. 4. The problem is reduced to the minimization of a function of 52 variables as in the plate example.

The load vs end-shortening plot is shown as the dashed line in Fig. 5 for the particular case when $A = B = 12$ in., $h = 0.1$ in., $\nu = 0.3$, $E = 30 \times 10^6$ psi and $r = 500$ in. The ordinate is the equivalent uniform edge load given by Eq. (34) with x replaced by s and y by $r\theta$. As the end-shortening δ is imposed, the load required to maintain equilibrium increases along the prebuckled portion of the curve to the point denoted by "a" in Fig. 6. When an attempt is made to induce additional end-shortening beyond this point, a point on the postbuckled portion of the curve is obtained. This is a manifestation of the "snap through" phenomenon. Also, when an attempt to generate a point on the postbuckled curve to the left of point b was made, a jump to the prebuckled curve resulted. It was not possible to obtain, by energy search, a curve connecting the prebuckled and postbuckled portions of the load vs end-shortening curve between points a and b. Displacement configurations of points on such a curve are not stable and therefore do not correspond to local minima of the potential energy.

The response of the transverse center deflection, w_0, is represented by the dashed curve in Fig. 6. With increasing compression, w_0 reaches a maximum positive value and then reverses. At point a in Fig. 6, which corresponds to point a of Fig. 5, the center deflection is negative. In the postbuckled region w_0 steadily decreases.

Curved Panel

This example is the same as the case of the slightly curved cylindrical shell panel except that the radius is 150 in. As the end-shortening δ was increased, starting from the origin in Fig. 5, points on the curve joining the origin and point 1 were obtained; when an attempt was made to impose an end-shortening beyond point 1, a jump occurred to the curve 2-3. An attempt to extend this curve to the left of point 2 resulted in the generation of a point on curve 0-1, whereas an effort to extend this curve to the right of point 3 gave a point on the curve 4-5. Point 4 is as far to the left as this branch of the curve could be extended without a jump to curve 2-3 occurring. Curves joining points 1-2 and 3-4 could not be generated since displacement configurations corresponding to points on these curves are unstable and therefore do not correspond to local minima of the potential energy function.

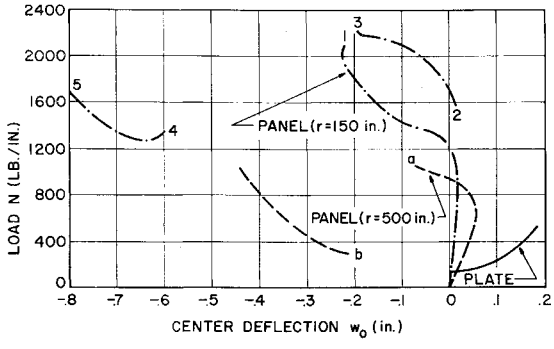


Fig. 6 Load vs center deflection.

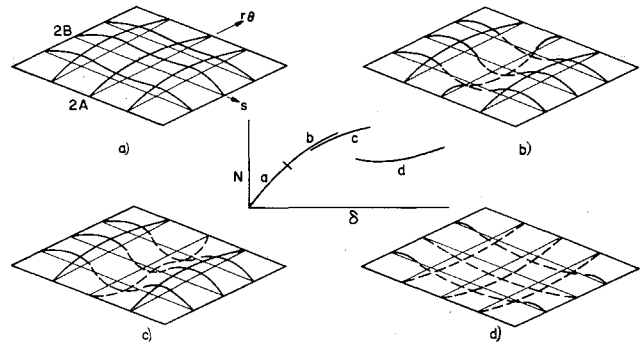


Fig. 7 Displacement configurations.

The response of the transverse center deflection w_0 is represented by the dash-dot curve in Fig. 6. The numbers 1, 2, 3, and 4 on the dash-dot curve in Fig. 6 correspond to the ends of the various portions of the load end-shortening curve indicated by the same number on the dash-dot curve in Fig. 5.

Some further insight into the behavior of this curved panel is offered by Fig. 7, which qualitatively illustrates displacement configurations corresponding to various portions of the load vs end-shortening curve. The displacement patterns shown in Figs. 7a and b are essentially the same except that in Fig. 7b the "dimpling inward" phenomenon has progressed to such a stage that the center deflection is negative. Figure 7c shows the displacement configuration after the first stage of the snap-through buckling of the panel manifested by a change in sign of the slope at the center of the edges $r\theta = \pm B$. In Fig. 7d the buckling pattern of the panel has undergone sign changes at all boundary points shown except at $(\pm A, \pm B/2)$. The displacement configuration illustrated in Fig. 7d suggests that another isolated portion of the load vs end-shortening curve may exist; however, the generation of the curve has not been carried to an extent beyond that shown.

Remarks of the Examples

The illustrative examples given are intended to indicate the applicability of the geometrically nonlinear plate and cylindrical-shell discrete elements presented to the prediction of finite displacements and postbuckling behavior. It is noted that since the possibility of antisymmetric behavior was precluded, the results presented may not represent the response that would have been predicted by modeling the entire structure instead of only a quadrant. Also, the four element modeling of a quadrant of the examples may not be sufficient to adequately describe the complicated displacement behavior of the cylindrical panel with larger curvature. A refinement of the modeling could lead to a better understanding of the load vs end-shortening curve corresponding to this example.

Concluding Remarks

An approach extending the discrete element displacement method to the nonlinear analysis of elastic structural systems that can be adequately idealized using rectangular plate and cylindrical shell elements has been presented. The formulation is such that numerical solutions can, by employing one of several powerful minimization algorithms, be obtained by seeking the minimum of the total potential energy.

The incorporation of geometric nonlinearity through the use of finite-displacement, strain-displacement relations results in fully coupled bending-membrane action. Further

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refinement could be achieved by employing more complete strain-displacement relations. It should be emphasized that the method presented does not require the use of load incrementation; however, the option to employ the method in conjunction with load incrementation or applied displacement incrementation exists.

The plate and shell discrete elements presented can be continuously joined together tangentially and at right angles through simple linking of nodal parameters. Extension to the case of plate and shell elements joined together continuously at any angle is possible; however, linking is no longer achieved by a simple one-to-one correspondence of variables.

The principal limitations of the method presented are: 1) it is limited to cases where the rotations of the deformed structure relative to the undeformed structure are small (i.e., $\sin\theta \approx \theta$, $\cos\theta \approx 1$); 2) it is limited to material behavior laws that can be represented by a strain energy density type of potential function; and 3) the use of assumed displacement states in the form of products of one-dimensional Hermite interpolation polynomials to facilitate the satisfaction of geometric admissibility conditions is limited to elements with boundaries parallel to orthogonal and skew coordinate systems.

Implicit in any discussion of the limitations of methods such as the one described in this Paper is the consideration of practicality in terms of computation time. In the present case, the typical solution times for a single load level (i.e., one point on the curve of Fig. 5) ranged between 5 and 10 min for the Fortran IV program used to generate the numerical results presented herein. It should further be noted that the computation times would be expected to rise quite rapidly as the problem size is increased. If compared with computation times for linear problems having a similar number of degrees of freedom these figures would be discouraging but in point of fact the present state of development of alternative generalized approaches to this class of problem admits no conclusive comparison.

On the other hand, the present capability possesses a considerable potential for improvement. A number of alternatives exist for the minimization section of the present code and if these can improve convergence or obviate some use of auxiliary storage, substantial savings will result. In addition, it may be possible to streamline the energy computation, which is the main time consumer.

The formulation lends itself to the consideration of a wide variety of alternative boundary conditions including various combinations of prescribed displacement and applied load distributions. The existing capability is thought to represent a useful tool for studying the finite deflection behavior of stiffened plate and shell structures including elastic buckling and postbuckling behavior.

Appendix A: Rigid-Body Modes

In order for a discrete element to represent the behavior of a portion of a structural system, it is necessary that the assumed displacement pattern be capable of approximating both rigid-body and deformation inducing displacements. In dealing with flat elements, it has been common to approximate the displacements with polynomial expansions which naturally contain the rigid-body modes and hence their representation has not presented a problem. However, in dealing with curved elements, polynomials can not in general be specialized to give the rigid-body modes. As pointed out in Ref. 17, it is not necessary that the element assumed displacement states represent the rigid-body displacement states exactly but rather the assumed displacement states must admit six linearly independent rigid-body modes in the limit as the element size is reduced. On the other hand, for an element to be useful, the rigid-body modes (as well as the deformation behavior) must be adequately approximated for elements large enough to limit the system degrees of freedom to a practical number.

The following remarks are intended to give some further insight into why the eigenvalue analysis of the 48 × 48 stiffness matrix indicated that the bicubic assumed displacement patterns admit six linearly independent displacement states which produce "very little" strain energy compared with the other displacement states even for relatively large-subtended angles.

For a specific set of strain-displacement relations, it will often be possible to determine the analytical form of six independent displacement states such that the strain energy in a discrete element is zero. For example, the following linear strain-displacement relations for a cylindrical shell element

$$\begin{aligned} \epsilon_s &= (\partial u/\partial s) - z(\partial^2 w/\partial s^2) \\ \epsilon_\theta &= \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{w}{r} - \frac{z}{r^2}\left(\frac{\partial^2 w}{\partial \theta^2} - \frac{\partial v}{\partial \theta}\right) \\ \gamma_{s\theta} &= \frac{\partial v}{\partial s} + \frac{1}{r}\frac{\partial u}{\partial \theta} - \frac{2z}{r}\left(\frac{\partial^2 w}{\partial s \partial \theta} - \frac{\partial v}{\partial s}\right) \end{aligned} \tag{A1}$$

imply the following zero strain-displacement states

$$\begin{aligned} u &= C_1 \sin\theta + C_2 \cos\theta + C_6 \\ v &= (s/r)(-C_1 \cos\theta + C_2 \sin\theta) + C_3 \sin\theta + C_4 \cos\theta + C_5 \\ w &= (s/r)(-C_1 \sin\theta - C_2 \cos\theta) - C_3 \cos\theta + C_4 \sin\theta \end{aligned} \tag{A2}$$

Sometimes the assumed displacement patterns are such that they can be specialized to represent the six zero strain-energy-displacement states exactly. In such cases it can be said that the rigid-body displacement states are represented exactly. It may be of interest, however, to compare the zero strain-energy-displacement states with the rigid-body displacement states of the element subjected to a general orthogonal transformation and a translation. These should agree, provided the basic simplifying assumptions made in the derivation of the strain-displacement relations are also invoked when comparing the general rigid-body displacement states with the zero strain-energy states. The basic criterion, however, is the strain energy.

Bicubic interpolation for all three displacements does not admit the six, linearly independent, strictly zero strain-energy-displacement states associated with the linear strain-displacement equations [Eqs. (A1)] and given by Eqs. (A2). An eigenvalue analysis of the associated stiffness matrix indicates that for practical subtended angles, there exist six linearly independent displacement states that produce very little strain energy compared with the other states.

Some qualitative insight into why the zero strain-displacement states [Eq. (A2)] are adequately approximated by the cubic interpolation is afforded by examining Taylor series expansion of $\sin\theta$ and $\cos\theta$. If the angle θ is small enough so that $\sin\theta$ may be replaced by $[\theta - (\theta^3/3!)]$, and $\cos\theta$ may be replaced by $[1 - (\theta^2/2!)]$ then the zero strain-energy-displacement states [Eqs. (A2)] may be expressed as follows:

$$\begin{aligned} u &\simeq C_1[\theta - (\theta^3/3!)] + C_2[1 - (\theta^2/2!)] + C_6 \\ v &\simeq \frac{s}{r}\left[-C_1\left(1 - \frac{\theta^2}{2!}\right) + C_2\left(\theta - \frac{\theta^3}{3!}\right)\right] + C_3\left(\theta - \frac{\theta^3}{3!}\right) + C_4\left(1 - \frac{\theta^2}{2!}\right) + C_5 \\ w &\simeq \frac{s}{r}\left[-C_1\left(\theta - \frac{\theta^3}{3!}\right) - C_2\left(1 - \frac{\theta^2}{2!}\right)\right] - C_3\left(1 - \frac{\theta^2}{2!}\right) + C_4\left(\theta - \frac{\theta^3}{3!}\right) \end{aligned} \tag{A3}$$

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and it is apparent that the bicubic interpolation for all displacements can be specialized to represent the displacement states exhibited in Eqs. (A3).

It is also interesting to note that the simplified linear strain displacement relations for a cylindrical shell

$$\begin{aligned} \epsilon_s &= (\partial u/\partial s) - z(\partial^2 w/\partial s^2) \\ \epsilon_\theta &= (1/r)(\partial v/\partial\theta) + (w/r) - (z/r^2)(\partial^2 w/\partial\theta^2) \\ \gamma_{s\theta} &= (\partial v/\partial s) + (1/r)(\partial u/\partial\theta) - (2z/r)(\partial^2 w/\partial s\partial\theta) \end{aligned} \quad \text{(A4)}$$

imply the following zero strain-energy-displacement states:

$$\begin{aligned} u &= C_1\theta - C_2(\theta^2/2) + C_6 \\ v &= (s/r)[-C_1 + C_2\theta] + C_3\theta - C_4(\theta^2/2) + C_5 \\ w &= -(s/r)C_2 - C_3 + C_4\theta \end{aligned} \quad \text{(A5)}$$

and it is apparent that the bicubic interpolation for all displacements can be specialized to represent the displacement states exhibited in Eqs. (A5).

Appendix B

The discretized strain energy for a typical element is obtained by substituting the assumed displacement modes [Eq. (3)] into Eqs. (8a–c) and (9a–c) and performing the indicated integrations over the middle surface of the element. The following notation is introduced so that the mathematical model can be expressed in relatively compact matrix form and to provide a convenient method for calculating an analytic gradient which is required for implementation of the function minimization technique. Define a vector X containing the 48 nodal degrees of freedom, partitioned as follows††:

$$\underset{48\times1}{X} = \begin{Bmatrix} \underset{16\times1}{\bar{U}} \\ \cdots \\ \underset{16\times1}{\bar{V}} \\ \cdots \\ \underset{16\times1}{\bar{W}} \end{Bmatrix} \quad \text{(B1)}$$

where

$$\bar{U}_p{}^T = [u_{11},u_{x11},u_{y11},u_{xy11},u_{12},u_{x12},u_{y12},u_{xy12},u_{22},u_{x22},u_{y22},u_{xy22}, u_{21},u_{x21},u_{y21},u_{xy21}] \quad \text{(B2)}$$

with similar definitions for the vectors $\bar{V}$ and $\bar{W}$. Also, define a vector $\bar{P}(x,y)$ the elements of which are the polynomial products in the assumed mode representation [Eq. (3)] that correspond to the elements of Eq. (B2); that is,

$$\underset{1\times16}{\bar{P}^T(x,y)} = [H_{01}{}^{(1)}(x)H_{01}{}^{(1)}(y),\ H_{11}{}^{(1)}(x)H_{01}{}^{(1)}(y),\ H_{01}{}^{(1)}(x)H_{11}{}^{(1)}(y),\ H_{11}{}^{(1)}(x)H_{11}{}^{(1)}(y), H_{01}{}^{(1)}(x)H_{02}{}^{(1)}(y), \ldots, H_{11}{}^{(1)}(x)H_{12}{}^{(1)}(y), H_{02}{}^{(1)}(x)H_{02}{}^{(1)}(y), \ldots, H_{12}{}^{(1)}(x)H_{12}{}^{(1)}(y), H_{02}{}^{(1)}(x)H_{01}{}^{(1)}(y), \ldots, H_{12}{}^{(1)}(x)H_{11}{}^{(1)}(y)] \quad \text{(B3)}$$

The displacement modes can now be written as the scalar product of two vectors,

$$\begin{aligned} u &= \bar{U}^T\bar{P} = \bar{P}^T\bar{U} \\ v &= \bar{V}^T\bar{P} = \bar{P}^T\bar{V} \\ w &= \bar{W}^T\bar{P} = \bar{P}^T\bar{W} \end{aligned} \quad \text{(B4)}$$

The polynomial vector $\bar{P}(x,y)$ is a function of the coordinates x and y, whereas the vectors of nodal variables $\bar{U},\bar{V}$, and $\bar{W}$ are constant with respect to these parameters; therefore, partial derivatives of the displacement functions can be written, for example,

$$\partial^2 w/\partial x\partial y = \bar{W}_p{}^T\bar{P}_{xy} = \bar{P}_{xy}{}^T\bar{W}_p$$

where the subscripts x and y denote partial differentiation.

A typical term from the quadratic strain energy [Eq. (8a)] is expressed, for example,

$$\frac{D}{2}\int_0^a\int_0^b \left(\frac{\partial^2 w}{\partial x^2}\right)^2 dydx = \frac{D}{2}\, W_p{}^T \times \left[\int_0^a\int_0^b \bar{P}_{xx}\bar{P}_{xx}{}^T dydx\right] W_p$$

Substitution of the displacement modes [Eq. (B4)] into the quadratic portion of the strain energy [Eq. (8a) or (9a)] and integrating results in the quadratic form

$$U_2 = \tfrac{1}{2}\ \underset{1\times48}{X^T}\ \underset{48\times48}{K_2}\ \underset{48\times1}{X} \quad \text{(B5)}$$

where the vector X is defined in Eq. (B1) and K_2 is the element stiffness matrix partitioned as follows:

$$\underset{48\times48}{K_2} = \begin{bmatrix} K_2{}^{(u)} & K_2{}^{(vu)T} & K_2{}^{(wu)T} \\ K_2{}^{(vu)} & K_2{}^{(v)} & K_2{}^{(wv)T} \\ K_2{}^{(wu)} & K_2{}^{(wv)} & K_2{}^{(w)} \end{bmatrix} \quad \text{(B6)}$$

where, for example (refer to Eq. 8a)

$$K_{2p}{}^{(u)} = \frac{12D}{h^2}\int_0^a\int_0^b \left[P_xP_x{}^T + \frac{1-\nu}{2}P_y\bar{P}_y{}^T\right] dydx$$

All of the submatrices are 16 × 16 dimensional arrays where those on the diagonal are symmetric and contain contributions that do not involve coupling between the u, v, and w displacements. The off-diagonal arrays contain contributions from the coupling terms as indicated by the superscripts and are not symmetric; the total element stiffness matrix K_2 is however, symmetric. Formulas for the elements of the submatrices of K_{2c}, the element stiffness matrix for the cylinder element, are given by Eqs. (12) of Ref. 12. Formulas for the elements of the submatrices of the plate element stiffness matrix K_{2p} are obtained from Eqs. (12) of Ref. 12 by replacing l with a, $r\Delta\theta$ with b and $1/r$ with zero.

In order to express the cubic and fourth-order contributions to the strain energy in matrix form, additional definitions are introduced. From Eq. (A4),

$$\partial w/\partial x = \bar{P}_x{}^T\bar{W}_p = \bar{P}_{1x}\bar{W}_{1p} + \bar{P}_{2x}\bar{W}_{2p} + \ldots + \bar{P}_{16x}\bar{W}_{16p}$$

where the numerical subscript denotes a particular element of the vectors $\bar{P}(x,y)$ and $\bar{W}_p$, and the subscript x denotes partial differentiation. Then

$$(\partial w/\partial x)^2 = \bar{P}_{1x}{}^2\bar{W}_{1p}{}^2 + 2\bar{P}_{1x}\bar{P}_{2x}\bar{W}_{1p}\bar{W}_{2p} + \bar{P}_{2x}\bar{W}_{2p}{}^2 + 2\bar{P}_{1x}\bar{P}_{3x}\bar{W}_{1p}\bar{W}_{3p} + 2\bar{P}_{2x}\bar{P}_{3x}\bar{W}_{2p}\bar{W}_{3p} + \bar{P}_{3x}{}^2\bar{W}_{3p}{}^1 + \ldots + \bar{P}_{16x}{}^2\bar{W}_{16p}{}^2 \quad \text{(B7)}$$

We would like to write this expression, which contains 16 × (16 + 1)/2 = 136 terms, as the scalar product of two vectors similar to Eq. (B4). Therefore, define the vector Y, which contains the 136 possible combinations of the 16 nodal variables contained in W [Eq. (B2)],

$$\underset{1\times136}{Y^T} = [\bar{W}_1{}^2,\ 2\bar{W}_1\bar{W}_2,\bar{W}_2{}^2,\ 2\bar{W}_1\bar{W}_3,2\bar{W}_2\bar{W}_3,\ \bar{W}_3{}^2,\ \ldots,\ \bar{W}_{16}{}^2] \quad \text{(B8)}$$

†† In the following development, quantities such as X which do not exhibit either the subscript p (plate) or c (cylinder) are understood to represent either case. Particular examples are given for the case of the plate element with the understanding that for the cylinder p is to be replaced by c, x by s, and y by $r\theta$.

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and the vector $Z^{(x)}(x,y)$ the elements of which are the polynomial products in Eq. (7) that correspond to the elements of Y [Eq. (A8)]; that is,

$$\underset{1\times 136}{Z^{(x)T}} = [\bar{P}_{1x}{}^2, \bar{P}_{1x}\bar{P}_{2x}{}^2, \bar{P}_{2x}, \bar{P}_{1x}\bar{P}_{3x}, \bar{P}_{2x}\bar{P}_{3x}, \bar{P}_{3x}{}^2, \ldots, \bar{P}_{16x}{}^2] \tag{B9}$$

Then,

$$(\partial w/\partial x)^2 = Y^T Z^{(x)} = Z^{(x)T} Y$$
$$(\partial w/\partial y)^2 = Y^T Z^{(y)} = Z^{(y)T} Y \tag{B10}$$

where $Z^{(y)}$ is of the same form as $Z^{(x)}$ and can be obtained by replacing x with y in Eq. (B9). A typical term from the cubic strain energy expression [Eq. (8b)] is then expressed, for example, as

$$\frac{D}{2}\int_0^a\int_0^b \frac{12}{h^2}\frac{\partial u}{\partial x}\left(\frac{\partial w}{\partial x}\right)^2 dy\,dx = \frac{12D}{2h^2} U^T\left[\int_0^a\int_0^b P_x Z^{(x)T}\,dy\,dx\right] Y$$

and an example from the fourth-order expression is

$$\frac{D}{2}\int_0^a\int_0^b \frac{12}{h^2}\frac{1}{4}\left(\frac{\partial w}{\partial x}\right)^4 dy\,dx = \frac{12D}{8h^2} Y^T\left[\int_0^a\int_0^b X^{(x)}Z^{(x)T}\,dy\,dx\right] Y$$

where Eqs. (B4) and (B10) were employed.

Substitution of Eqs. (B4) and (B10) into the cubic and fourth-order strain-energy terms [Eqs. (8b, c, and 9b, c)] and integrating, results in the matrix equations

$$U_3 = \underset{1\times 48}{\tfrac{1}{2}\,X^T}\ \underset{48\times 138}{K_3}\ \underset{136\times 136}{Y} \tag{B11}$$

and

$$U_4 = \underset{1\times 136}{\tfrac{1}{8}\,Y^T}\ \underset{136\times 136}{K_4}\ \underset{136\times 1}{Y} \tag{B12}$$

The matrix K_4 contains contributions from terms involving only the transverse displacement w; however, K_3 contains terms in u, v, and w which are coupled and is therefore partitioned as follows:

$$\underset{48\times 136}{K_3} = \begin{Bmatrix} K_3{}^{(uw)} \\ \cdots \\ K_3{}^{(vw)} \\ \cdots \\ K_3{}^{(ww)} \end{Bmatrix} \tag{B13}$$

where each of the submatrices are of dimension 16×136 and the superscripts denote particular variable coupling. Since there are no terms in U_{3p} that involve only w contributions, it follows that $K_{3p}{}^{(ww)} \equiv 0$; however the corresponding submatrix for the cylindrical element $K_{3c}{}^{(ww)}$ is not zero as can be seen from Eq. (9b).

Analytic Gradient

In order to exploit to full advantage the convergence and stability properties of the function minimization technique which is employed[15] to locate the unconstrained minimum of the total potential energy for an assemblage of discrete elements, it is desirable to possess the capability for calculating an analytic gradient of the strain-energy expression. The matrix representation for the discretized strain energy of a typical element developed in the foregoing discussion [Eqs. (B5, B11, and B12)], provides the basis for expressing the gradient in corresponding matrix notation with relative ease.

The gradient of the quadratic contribution to the element strain energy [Eq. (B5)] is expressed by (since the stiffness matrix K_2 is symmetric)

$$\underset{48\times 1}{G_2(X)} = \underset{48\times 48}{K_2}\ \underset{48\times 1}{X} \tag{B14}$$

The gradient of the fourth-order contribution [Eq. (B12)] with respect to the components of the vector Y [Eq. (B8)] is given similarly by

$$\underset{136\times 1}{G_{4y}(Y)} = \underset{136\times 136}{\tfrac{1}{4}\,K_4}\ \underset{136\times 1}{Y} \tag{B15}$$

However, it is desired to obtain the gradient of U_4 with respect to the components of the vector $\bar{W}$ [Eq. (B2)]. This can be done as follows: write

$$\underset{16\times 1}{G_{4w}} = \begin{Bmatrix} \partial U_4/\bar{W}_1 \\ \cdot \\ \cdot \\ \cdot \\ \partial U_4/\bar{W}_{16} \end{Bmatrix}$$

or employing the chain rule of differentiation,

$$G_{4w} = \begin{Bmatrix} \dfrac{\partial U_4}{\partial Y_1}\dfrac{\partial Y_1}{\partial \bar{W}_1} + \cdots + \dfrac{\partial U_4}{\partial Y_{136}}\dfrac{\partial Y_{136}}{\partial \bar{W}_1} \\ \cdot \\ \cdot \\ \cdot \\ \dfrac{\partial U_4}{\partial Y_1}\dfrac{\partial Y_1}{\partial \bar{W}_{16}} + \cdots + \dfrac{\partial U_4}{\partial Y_{136}}\dfrac{\partial Y_{136}}{\partial \bar{W}_{16}} \end{Bmatrix} \tag{B16}$$

Now, define the matrix J of partial derivatives,

$$\underset{15\times 136}{J} = \begin{bmatrix} \dfrac{\partial Y_1}{\partial \bar{W}_1} & \dfrac{\partial Y_2}{\partial \bar{W}_1} & \cdots & \dfrac{\partial Y_{136}}{\partial \bar{W}_1} \\ \dfrac{\partial Y_1}{\partial \bar{W}_2} & \dfrac{\partial Y_2}{\partial \bar{W}_2} & \cdots & \dfrac{\partial Y_{136}}{\partial \bar{W}_2} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & & \cdot \\ \dfrac{\partial Y_1}{\partial \bar{W}_{16}} & \dfrac{\partial Y_2}{\partial \bar{W}_{18}} & \cdots & \dfrac{\partial Y_{136}}{\partial \bar{W}_{16}} \end{bmatrix} \tag{B17}$$

Then Eq. (B16) can be written as

$$\underset{16\times 1}{G_{4w}} = \underset{16\times 136}{J}\ \underset{136\times 1}{G_{4y}}$$

or, finally, from Eq. (B15),

$$\underset{16\times 1}{G_{4w}} = \underset{16\times 136}{\tfrac{1}{4}\,J}\ \underset{136\times 136}{K_4}\ \underset{136\times 1}{Y} \tag{B18}$$

The gradient of the cubic contribution [Eq. (B11)] to the element strain energy is obtained in two parts:

$$\underset{48\times 1}{G_{3x}} = \underset{48\times 136}{\tfrac{1}{2}\,K_3}\ \underset{136\times 1}{Y} \tag{B19}$$

and,

$$\underset{16\times 1}{G_{3w}} = \underset{16\times 136}{\tfrac{1}{2}\,J}\ \underset{136\times 48}{K_3{}^T}\ \underset{48\times 1}{X} \tag{B20}$$

where J is defined by Eq. (B17).

With the notation of Eqs. (B14, B18, B19, and B20), the analytic gradient of the total element strain energy is expressed by

$$\underset{48\times 1}{\nabla U} = \underset{48\times 1}{G_2} + \underset{48\times 1}{G_{3w}} + \underset{48\times 1}{\tilde{G}} \tag{B21}$$

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where,

$$\underset{48\times1}{\tilde{G}} = \begin{Bmatrix} \underset{16\times1}{0} \\ \cdots \\ \underset{16\times1}{0} \\ \cdots \\ \underset{16\times1}{G_{4w} + G_{3w}} \end{Bmatrix}$$

Also, the gradient of the work term [Eq. (12)] is

$$\underset{48\times1}{\nabla W} = \underset{48\times1}{P} \tag{B22}$$

where P is the work equivalent load vector. The gradient of the total potential energy for a discrete element is then expressed by

$$\nabla \pi_p = \nabla U - \nabla W \tag{B23}$$

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